TT-W-535C
November 28, 1975
SUPERSEDING
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May 12, 1967

FEDERAL SPECIFICATION

WOOD PRESERVATIVE: FLUOR CHROME ARSENATE PHENOL MIXTURE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the composition and methods of analysis of a fluor chrome arsenate phenol treating solution for the preservative treatment of wood according to TT-W-571 (see 6.1), and also a strong solution suitable for the brush treatment of the cut ends of treated wood products.

1.2 Classification.

1.2.1 Composition. Fluor chrome arsenate phenol shall be of one composition, as specified herein.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-W-571 - Wood Preservation: Treating Practices.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements, as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

40 CFR 162 - Federal Insecticide, Fungicide, and Rodenticide Act.
49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Wood-Preservers' Association (AWPA) Standard:

A2 - Standard Methods for Analysis of Water-Borne Preservatives and Fire-Retardant Formulation.

(Application for copies should be addressed to the American Wood-Preservers' Association, 1625 I Street, NW, Washington, DC 20006.)

3. REQUIREMENTS

3.1 Properties.

3.1.1 Fluor chrome arsenate phenol mixture in solid form shall consist of water-soluble compounds selected from the following groups, each in excess of 95 percent purity on an air-dry basis:

Fluoride - e.g., sodium fluoride
Pentavalent arsenate - e.g., disodium arsenate or arsenic pentoxide
Hexavalent chromium - e.g., sodium chromate, sodium dichromate, or potassium dichromate
Dinitrophenol or sodium pentachlorophenoxide

The commercial preservative shall be labeled as to its total content of active ingredients calculated as fluoride (F), arsenic pentoxide (As_2O_5), chromium trioxide (CrO_3), and dinitrophenol ($(NO_2)_2C_6H_3OH$) or sodium pentachlorophenoxide ($NaOC_6Cl_5$).

3.1.2 The active ingredients of fluor chrome arsenate phenol, either in solid form, in a treating solution, or in a concentrated solution, shall be present in the following ratios:

Fluoride, as F	22 parts by weight
Hexavalent chromium, as CrO_3	37 parts by weight
Pentavalent arsenic, as As_2O_5	25 parts by weight
Dinitrophenol or sodium pentachlorophenoxide	16 parts by weight

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3.1.3 The composition of the solid preservative, or of a treating solution, or of a concentrated solution, may vary within the following limits:

	Minimum percent	Maximum percent
Fluoride, as F	20	24
Hexavalent chromium, as CrO_3	33	41
Arsenic, as As_2O_5	22	28
Dinitrophenol or sodium pentachlorophenoxide	14	18

3.1.4 The pH of the treating solution shall be maintained between 5.5 and 7.8. Sodium or potassium hydroxide may be used to adjust the pH. A solution of the preservative shall be essentially free of insoluble matter.

3.1.5 The concentration of a treating solution shall be adjusted so as to result in a retention as specified for the product. A strong solution shall contain a minimum of 5 percent fluor chrome arsenate phenol.

3.1.6 The net retention of preservative shall be calculated as CrO_3 plus As_2O_5 deposited in the wood in the proportions as limited in 3.1.3. The total weight of CrO_3 plus As_2O_5 shall be multiplied by a factor of 1.6 to correct for active ingredients not determined by chemical analysis.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The treating solution shall be thoroughly agitated, after which a 1-quart sample shall be taken for inspection and test purposes. The contractor shall supply such samples and containers as may be required, at no cost to the Government.

4.3 Tests. Tests to determine the properties of fluor chrome arsenate phenol treating solution (see 3.1) shall be made in accordance with AWPA Standard A2.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A or B. The preservative shall be furnished in 5-gallon pails conforming to type I, class 2, 3, 4, or 5 of PPP-P-704 or 55-gallon steel drums conforming to type I or II of PPP-D-729. Five-gallon containers shall be furnished with pour spouts (see 6.2).

5.1.2 Level C. The preservative shall be furnished in 5-gallon pails or in 55-gallon drums normally furnished for commercial practice (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Fluor chrome arsenate phenol mixture may be used for the preservative treatment of lumber, structural timbers, mine timbers, and plywood in accordance with the conditions outlined in Federal Specification TT-W-571, Wood Preservation: Treating Practices. A strong solution (see 3.1.5) may be used for brush applications to cut surfaces of treated products.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable level of packing required (see 5.1).
(c) Size of container required for level A, B, or C packing (see 5.1.1 and 5.1.2).

6.3 Normally, only treated materials will be procured by the Government, and there are no Government plants using the bulk preservatives.

6.3.1 If the material covered by this specification is procured by the Government as an end item, it should be marked in accordance with the Federal Insecticide, Fungicide, and Rodenticide Act (see 2.1).

6.4 The specification was developed by the Forest Products Laboratory, Forest Service, U.S. Department of Agriculture, Madison, Wisconsin, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

Preparing activity:

AGR - FS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 25 cents each.

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